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Multiple imputation with missing data indicators

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Abstract

Multiple imputation is a well-established general technique for analyzing data with missing values. A convenient way to implement multiple imputation is sequential regression multiple imputation, also called chained equations multiple imputation. In this approach, we impute missing values using regression models for each variable, conditional on the other variables in the data. This approach, however, assumes that the missingness mechanism is missing at random, and it is not well-justified under not-at-random missingness without additional modification. In this paper, we describe how we can generalize the sequential regression multiple imputation imputation procedure to handle missingness not at random in the setting where missingness may depend on other variables that are also missing but not on the missing variable itself, conditioning on fully observed variables. We provide algebraic justification for several generalizations of standard sequential regression multiple imputation using Taylor series and other approximations of the target imputation distribution under missingness not at random. Resulting regression model approximations include indicators for missingness, interactions, or other functions of the missingness not at random missingness model and observed data. In a simulation study, we demonstrate that the proposed sequential regression multiple imputation modifications result in reduced bias in the final analysis compared to standard sequential regression multiple imputation, with an approximation strategy involving inclusion of an offset in the imputation model performing the best overall. The method is illustrated in a breast cancer study, where the goal is to estimate the prevalence of a specific genetic pathogenic variant.

Keywords

Chained equations multiple imputation, not missing at random, missing data indicator, sequential regression multiple imputation

Introduction

Multiple imputation has become a popular and effective approach for analyzing datasets with missing values.[1–3] This general approach relies on an assumed statistical model for the variables with missing values. If this model is appropriately specified and the mechanism generating missingness in the data depends only on fully observed data (called missing at random [MAR]), then this method has been shown to have good theoretical and numerical properties.[4] In analyzing data in practice, analysts must make good choices in specifying models used for imputation, and they must determine whether the MAR missingness assumption is plausible or at least approximately satisfied.

When missingness depends on unobserved data conditional on the observed data, called missing not at random (MNAR), then many standard multiple imputation strategies cannot be directly applied.[2] For example, suppose we have three variables in our data (denoted X_1, X_2, and X_3) and that X_1 and X_2 have missing values for some subjects. Let R_j be the indicator of whether X_j is observed ($R_j = 1$) or not ($R_j = 0$). Missingness in X_1 is MAR if $P(R_1 = 1|X)$ depends only on X_3. Missingness is MNAR if missingness in X_1 depends directly on the value of X_1 or if it depends on

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X_2, which is also sometimes missing. If we were to impute missing values of X_1 and X_2 ignoring MNAR missingness, we may introduce bias in estimating parameters of interest later on.

It is well-known that it is impossible to distinguish between MNAR and MAR missingness using the observed data alone.[5] Therefore, a general recommendation is to use a large number of observed variables to impute the missing data, since it may be more reasonable to assume MAR missingness when we condition on a larger amount of the observed data. Another general approach is to perform a sensitivity analysis exploring how much final analysis conclusions are impacted when we perform imputation from distributions incorporating different plausible MNAR assumptions (i.e. models for R_1 and R_2 with corresponding fixed parameter values). These imputation distributions, however, can often be complicated functions of the data models (models for X) and the assumed models for missingness (model for $R|X$). Approximations of these imputation distributions can provide an easier path toward routine implementation.

The ideal way to impute variables with missing values under MAR is to specify a joint distribution for all the X variables and then use the conditional distribution derived from that joint distribution to impute missing values. It is challenging to specify such a joint distribution when many variables have missing values and the variables may be of mixed types, such as binary, categorical and continuous. A convenient and pragmatic way to overcome this problem is to perform chained equations multiple imputation, also known as sequential regression multiple imputation [denoted SRMI].[6–8,3] This method is also referred to in the statistical literature as multiple imputation by chained equations (MICE) and fully conditional specification (FCS). In this approach, a regression model is specified for imputing each variable with missing values, conditional on all the other variables. The variables with missing values in X are then imputed sequentially, and the procedure is iterated a few times until stable results are obtained.

SRMI can be thought of as mimicking an iterative Markov chain Monte Carlo (MCMC) algorithm under a full Bayesian joint model with flat priors, where missing values are viewed as parameters and are drawn from corresponding posterior distributions. The posterior distribution for imputing each variable is the conditional distribution of that variable given all the others, which is analogous to the SRMI approach. The SRMI standard practice for sequentially imputing variables in X conditional on the other X variables can be extended to also condition on response indicators, R_1 and R_2. As a generalization of SRMI under MNAR missingness, some researchers propose including missingness indicators R_1 and R_2 as predictors in regression models used for imputation.[9,10] When imputing missing values of X_1, for example, we might include R_2 as a covariate in the model that is used for imputation. R_1 may also be incorporated into the imputation of X_1 through a corresponding fixed parameter, δ, used in sensitivity analysis to control the degree of MNAR dependence between X_1 and R_1. However, it is unclear how well these strategies approximate the true posterior distribution and in what settings this approach is justified.

In this paper, we primarily explore a particular missingness scenario where missingness in each covariate is MNAR dependent on *other* variables that themselves have missing values but where does not depend on the value of the missing covariate itself, conditional on fully observed variables. This setting may occur, for example, if decisions for whether or not a medical test is performed are based on other incompletely recorded patient characteristics. In this missingness setting, we derive regression model approximations for imputing normally distributed, binary, and categorical variables within the SRMI algorithm under this form of MNAR. This work provides theoretical justification for existing modifications of the SRMI procedure under MNAR and suggests several new extensions that may outperform existing SRMI strategies in certain settings. The paper is organized as follows: we first propose extensions of SRMI for handling MNAR missingness, including an exact imputation strategy and several simple approximations. We then compare the performance of these different approximation strategies in terms of bias in estimating downstream regression model parameters in a simulation study. We then apply these methods to handle informative missingness in a motivating study of the prevalence of BRCA1 and BRCA2 pathogenic variants among women newly-diagnosed with breast cancer, where missingness in the BRCA1/2 status is likely related to familial history of breast cancer diagnosis, which is also only partially observed. Finally, we present a discussion.

Sequential regression multiple imputation under MNAR

Deriving the conditional imputation distribution

Assume we have a dataset consisting of n independent observations in p variables, denoted $X_1, \ldots, X_p$. For each subject, let $R_j = 0$ if X_j is missing and $R_j = 1$ if X_j is observed. Let $X_{(-j)}$ denote the $p-1$ variables in X left after excluding X_j, and let $R_{(-j)}$ denote the $p-1$ variables in R left after excluding R_j. To avoid the situation where an observation has missing values for all X's, we will assume that at least one of the X's has no missing values for every subject. We will also assume a non-monotone pattern of missingness, by which we mean there is no (j, k) pair of variables for which $R_j = 0$

implies $R_k = 0$. Our target of interest is some aspect of the joint distribution of $X_1, \ldots, X_p$, such as the coefficients in the regression model of X_1 on all the other X's or the mean of X_1.

We propose using a sequential regression multiple imputation (SRMI) scheme to obtain B complete datasets with the missing X's filled in. We then follow the standard approach[1] of analyzing each imputed dataset separately with the desired model and then combining those results to give final estimates and confidence intervals. We want to impute each variable X_j with missing values from its assumed distribution given $X_{(-j)}$ and R, denoted $f(X_j|X_{(-j)}, R)$. Some form of regression model can be used to approximate this distribution, where each regression model is tailored to the variable type for X_j, for example, logistic regression if X_j is binary, linear regression if X_j is continuous, etc. In practice, these regression models are usually specified to have a linear combination of the variables on the right hand side, but these models could also be more flexible and include non-linear and interaction terms. The question then becomes how R should be incorporated into the imputation regression models. One strategy is to include $R_{(-j)}$ directly as additional predictors in the imputation model. Since we cannot use the observed data to reliably estimate the association between X_j and R_j, R_j can be indirectly incorporated into the imputation regression model through a fixed offset term $\delta_j R_j$, where δ_j is treated as a sensitivity analysis parameter. Mercaldo and Blume[10] called this strategy multiple imputation with missing indicators (MIMI), and Tompsett and White[9] also advocates for its general use. We will call this general strategy "sequential regression multiple imputation with missing indicators," denoted SRMI-MI, and we focus on the particular setting where $\delta_j = 0$ under Assumptions 1 and 2 below.

One justification for including the extra terms $R_{(-j)}$ in the imputation models is simply as a way to make the imputation model more flexible and allow the whole imputation procedure to be less reliant on the possibly restrictive assumptions of imputation models with small numbers of parameters. A more formal justification can be obtained by considering a Bayesian MCMC approach for the problem. Mimicking the ideas developed for other models,[11,12] we obtain the form of the ideal conditional distribution expressed such that the imputation distribution is congenial with, or at least approximately congenial with, the desired target model of the analyst.[13] Suppose that the desired target analysis model is some function of the joint distribution of $X_1, \ldots, X_p$, written as $f(X_1, \ldots, X_p)$. This joint distribution would then determine the form of any submodel based on X, such as the marginal distribution of X_j or the conditional distribution of $X_j|X_{(-j)}$. Treating $(R_1, \ldots, R_p)$ as random variables, we write the joint distribution of $X_1, \ldots, X_p, R_1, \ldots, R_p$ as $f(X_1, \ldots, X_p, R_1, \ldots, R_p)$, which can be factored in a selection model form as

$$f(X_1, \ldots, X_p) \times f(R_1, \ldots, R_p|X_1, \ldots, X_p)$$

In the MCMC algorithm, we would ideally draw missing values of X_j from the following conditional distribution:

$$f(X_j|X_{(-j)}, R) \propto f(X_j|X_{(-j)}) f(R_j|X, R_{(-j)}) f(R_{(-j)}|X) \tag{1}$$

viewed as a function of X_j. In this expression, the distribution $f(R_j|X, R_{(-j)})$ is not identified using the observed data, and $f(R_{(-j)}|X)$ may take a complicated form in general. In order to focus our attention on a more tractable missing data setting, we make the following two assumptions:

Assumption 1 The R_j's are conditionally independent given $X_1, \ldots, X_p$.
Assumption 2 The missingness in X_j does not depend on X_j, that is, $f(R_j|X) = f(R_j|X_{(-j)})$.

The second assumption allows the missingness of one variable R_j to depend on another variable X_k, $k \neq j$, which itself may be missing. In this sense, this setting is a relaxation of the usual missing at random assumption, where missingness may depend only on variables that are fully observed given the observed data. We view the first assumption as a mild one, and it could be relaxed to have blocks of R_j's be conditionally independent. The second assumption is a stronger one, and its reasonableness will depend on context of the missing data problem. For a recent work on handling missingness dependent on a covariate's own values, see Beesley and Taylor (2021).[14] Under Assumptions 1 and 2, we can simplify equation (1) as follows:

$$f(X_j|X_{(-j)}, R) \propto f(X_j|X_{(-j)}) \prod_{k \neq j} f(R_k|X_j, X_{(-j)}) \tag{2}$$

We see immediately that R_j does not occur in this expression. Additionally, any missingness indicator R_k such that $R_k \perp X_j|X_{(-j)}$ can also be ignored. The imputation distribution of X_j, therefore, will depend on $X_{(-j)}$ and any indicator R_k, $k \neq j$ such that $R_k \perp X_j|X_{(-j)}$. The distribution in equation (2) will generally be a messy expression. We can apply importance sampling methods, rejection sampling, weighting, Metropolis-Hastings algorithms, or grid-based sampling to draw directly from equation (2). In the case of rejection sampling, for example, we could draw candidate imputations from $f(X_j|X_{(-j)})$ and

accept the first candidate draw that satisfies $U < \prod_{k \neq j} f(R_k | X_j, X_{(-j)})$, where random variable U is drawn from a uniform(0,1) distribution and the missingness models densities are evaluated at draws of the corresponding model parameters (see Supplemental Section D for details). When X_j is categorical, the exact form of the probability mass function can be worked out based on equation (2) as in equation (3). In general, we may not want to specify parametric models for the missingness probabilities, or we may prefer to impute using regression model structures. In the remainder of this paper, we will consider approximations to equation (2) that could be more easily implemented in a SRMI-MI algorithm.

A subtle but noteworthy issue is that the distribution in equation (2) does not condition on model parameters and is instead only a function of the data. For multiple imputation, drawing from the conditional distribution without parameters is usually achieved in two stages, first by drawing parameters of the model and then imputing the variable from the conditional distribution based on that parameter value. The same technique would be used for equation (2), in which parameters for the component distributions are *drawn* from distributions that are derived using the available data. This is often implemented by fitting the corresponding component model on a bootstrap sample of the data or by making a multivariate normal approximation.[2] The question then becomes which subset of the data should be used to derive the distribution from which to perform these parameter draws. In a Bayesian MCMC algorithm, parameters are drawn conditional on the most recently drawn values for all other parameters. In the missing data setting, this would suggest drawing model parameters using the most recently imputed data for the *entire* dataset of size n. In contrast, usual implementation of SRMI methods draw imputation model parameters for imputing X_j using the data with X_j observed, here called local complete case data, and treating the most recent imputations of $X_{(-j)}$ as if they were observed. It is feasible to adapt SRMI to make use of all n observations, that is, use current imputed values of X_j and $X_{(-j)}$ in the estimation of the regression model for X_j given all other variables. It is a easy to show that, theoretically, either approach can be used when missingness in X_j is independent of R_j. In practice, imputing within SRMI based on all n observations may be preferred simply because the increased sample size may give better estimates of the relationship between each X_j and $R_{(-j)}$. This approach is used in our simulations and data analysis.

Regression model approximations for imputing binary, categorical and continuous variables

In this section, we approximate the imputation distribution proportional to equation (2) under different assumptions about the distributions of the variables in X.

Imputing binary variables

Suppose we want to impute binary variable X_1 and that the distribution for $X_1 | X_{(-1)}$ is well-approximated by a logistic regression model as follows:

$$P1 = P(X_1 = 1 | X_{(-1)}) = \text{expit}(\theta_0 + \Sigma_{j=2}^{p} \theta_j X_j)$$

where $\text{expit}(u) = \exp(u)/(1 + \exp(u))$. Let $PRj(x_1)$ denote the probability of observing X_j given $X_{(-j)}$ with $X_1 = x_1$. We note that $PRj(x_1)$ can be a function of all the X's except X_j, but for convenience we use the notation $PRj(x_1)$. Thus, for example, $PR2(1) = P(R_2 = 1 | X_1 = 1, X_3, , X_p)$. Following this notation and accounting for proportionality, we can express equation (2) as $P(X_1 = 1 | X_{(-1)}, R) = A/(A + B)$ where

$$A = P1 \prod_{j=2}^{p} PRj(1)^{R_j} \left[1 - PRj(1)\right]^{1-R_j}$$

$$\text{and } B = (1 - P1) \prod_{j=2}^{p} PRj(0)^{R_j} \left[1 - PRj(0)\right]^{1-R_j}$$

This expression simplifies as follows:

$$\begin{aligned} &\log\left[\frac{P(X_1 = 1 | X_{(-1)}, R)}{1 - P(X_1 = 1 | X_{(-1)}, R)}\right] \\ &= \log\left[\frac{P1}{1 - P1}\right] + \sum_{j=2}^{p} \left\{ R_j \log\left[\frac{PRj(1)}{PRj(0)}\right] + (1 - R_j) \log\left[\frac{1 - PRj(1)}{1 - PRj(0)}\right] \right\} \\ &= \theta_0 + \sum_{j=2}^{p} \theta_j X_j + \sum_{j=2}^{p} \left\{ R_j \log\left[\frac{PRj(1)}{PRj(0)}\right] + (1 - R_j) \log\left[\frac{1 - PRj(1)}{1 - PRj(0)}\right] \right\} \end{aligned} \tag{3}$$

This can also be rewritten as

$$\log\left[\frac{P(X_1 = 1|X_{(-1)}, R)}{1 - P(X_1 = 1|X_{(-1)}, R)}\right]$$
$$= \theta_0 + \sum_{j=2}^{p} \theta_j X_j + \sum_{j=2}^{p} R_j \log\left[\frac{PRj(1)}{PRj(0)}\frac{\{1 - PRj(0)\}}{\{1 - PRj(1)\}}\right] + \sum_{j=2}^{p} \log\left[\frac{1 - PRj(1)}{1 - PRj(0)}\right]$$

We now consider several special cases and then propose a general strategy for imputation of a binary variable.

Binary Special Case 1: logistic missingness with main effects.

Suppose that the model for missingness for each variable X_j can be expressed as follows:

$$PRj(X_1) = P(R_j = 1|X_1, X_{(-1)}) = \text{expit}(\phi_{j0} + \Sigma_{k\neq j}\phi_{jk}X_k) \tag{4}$$

In this case, equation (3) can be simplified as

$$\text{logit}[P(X_1 = 1|X_{(-1)}, R)] = \theta_0 + \sum_{j=2}^{p} \theta_j X_j + \sum_{j=2}^{p} \phi_{j1} R_j$$
$$+ \sum_{j=2}^{p} \log\left[1 + \exp\left(\phi_{j0} + \sum_{k=2,k\neq j}^{p} \phi_{jk}X_k\right)\right] - \log\left[1 + \exp\left(\phi_{j0} + \phi_{j1} + \sum_{k=2,k\neq j}^{p} \phi_{jk}X_k\right)\right]$$

In the special case where $p = 3$ and X_2 and X_3 are binary, all the terms involving the *log*'s can be simplified and combined with θ_0 and the $\theta_j X_j$'s, and the final expression is simply a linear combination of $X_2, , X_p$ and $R_2, , R_p$ as follows:

$$\text{logit}\big[P(X_1 = 1|X_{(-1)}, R)\big] = \omega_0 + \sum_{j=2}^{p} \omega_j X_j + \sum_{j=2}^{p} \omega_{Rj} R_j \tag{5}$$

In general for $p > 3$ and for non-binary X_2 and X_3, equation (3) does not reduce to this simple additive form. However, a first order Taylor series approximation of the logarithm terms (assuming all values of ϕ_{jk}, $k \neq j$ are small) does lead to equation (5) as an approximation to the desired imputation distribution. A second order Taylor series approximation results in the following regression model structure:

$$\text{logit}\big[P(X_1 = 1|X_{(-1)}, R)\big] \approx \alpha_0 + \sum_{k=2}^{p} \alpha_k X_k + \sum_{k=2}^{p} \alpha_{Rk} R_k + \sum_{j=2}^{p}\sum_{k=2}^{p} \alpha_{2jk} X_j X_k \tag{6}$$

to impute X_1, that is, including interactions between the X's.

Binary Special Case 2: interactions in logistic missingness model.

Suppose instead that the missingness models include interactions between other covariates and X_1. For simplicity, we will assume $p = 3$. Suppose that

$$\text{logit}\big[PR2(X_1)\big] = \phi_{20} + \phi_{21}X_1 + \phi_{23}X_3 + \phi_{24}X_1X_3$$
$$\text{logit}\big[PR3(X_1)\big] = \phi_{30} + \phi_{31}X_1 + \phi_{32}X_2 + \phi_{34}X_1X_2$$

In this case, the imputation takes the following form:

$$\text{logit}\big[P(X_1 = 1|X_2, X_3, R_2, R_3)\big] = \theta_0 + \theta_2 X_2 + \theta_3 X_3$$
$$+ R_2\big[\phi_{21} + \phi_{24}X_3\big] + R_3\big[\phi_{31} + \phi_{34}X_2\big]$$
$$+ \log\big[1 + \exp(\phi_{20} + \phi_{23}X_3)\big] - \log\big[1 + \exp(\phi_{20} + \phi_{21} + \phi_{23}X_3 + \phi_{24}X_3)\big]$$
$$+ \log\big[1 + \exp(\phi_{30} + \phi_{32}X_2)\big] - \log\big[1 + \exp(\phi_{30} + \phi_{31} + \phi_{32}X_2 + \phi_{34}X_2)\big]$$

Using the same logic as before and applying a first-order Taylor series approximation, we can express the imputation distribution as follows:

$$\begin{aligned}&\text{logit}\big[P(X_1 = 1|X_2, X_3, R_2, R_3)\big]\\&\approx \omega_0 + \omega_2 X_2 + \omega_3 X_3 + \omega_{R2} R_2 + \omega_{R3} R_3 + \omega_{3,R2} X_3 R_2 + \omega_{2,R3} X_2 R_3\end{aligned} \tag{7}$$

For $p > 3$, we can similarly approximate the imputation distributions by including interactions between the X's and missingness indicators.

Binary General Case.

Suppose now that variables $X_2, , X_p$ have some unspecified form and we allow $PRj(X_1) = P(R_j = 1|X)$ to take more general (e.g. non-logistic) form. We notice that equation (3) resembles a logistic regression model with predictors $X_{(-1)}$ and a term that is a function of the missingness indicators, $R_{(-j)}$, and the probabilities of missingness, $PRj(X_1)$. Guided by equation (3), we propose the following strategy for imputing missing values of X_1 within each iteration of a chained equations imputation algorithm:

1. For each $j > 1$, fit a model (e.g. logistic or probit regression or even a regression tree) to the current imputed dataset of size n for the probability that X_j is observed.
2. For each observation and each $j > 1$, use these model estimates to calculate the probability that X_j is observed with X_1 set to 0 and with X_1 set to 1 to give $PRj(0)$ and $PRj(1)$, respectively. To calculate these probabilities, use the most recent imputed values for $X_{(-j)}$.
3. Define new variables

$$Z_j = R_j \log\left[\frac{PRj(1)}{PRj(0)}\right] + (1 - R_j)\log\left[\frac{1 - PRj(1)}{1 - PRj(0)}\right] \tag{8}$$

4. Impute X_1 using the following model:

$$\text{logit}\big[P(X_1 = 1|X_{(-1)}, R_2, R_3, Z_2, Z_3)\big] = \omega_0 + \sum_{k=2}^{p} \omega_k X_k + \sum_{k=2}^{p} Z_j \tag{9}$$

where the ω's are first drawn from an approximation to their posterior distribution derived from a model fit to the full imputed dataset and where $\sum_{k=2}^{p} Z_j$ is a fixed offset (with coefficient equal to 1).

Imputing multinomial variables

Now, we suppose that X_1 is a categorical variable taking values in 0, 1, ..., S and that the distribution for $X_1|X_{(-1)}$ is well-approximated by a multinomial regression as follows:

$$PS = P(X_1 = s|X_{(-1)}) = \frac{\exp\left(\theta_{0s} + \sum_{j=2}^{p} \theta_{js} X_j\right)}{1 + \sum_{r=1}^{S} \exp\left(\theta_{0r} + \sum_{j=2}^{p} \theta_{jr} X_j\right)}$$

where all θ_{j0}'s are equal to zero. As in the derivation of equation (3), we can write the imputation distribution as follows:

$$\begin{aligned}&\log\left[\frac{P(X_1 = s|X_{(-1)}, R)}{P(X_1 = 0|X_{(-1)}, R)}\right]\\&= \theta_{0s} + \sum_{j=2}^{p} \theta_{js} X_j + \sum_{j=2}^{p}\left\{R_j \log\left[\frac{PRj(s)}{PRj(0)}\right] + (1 - R_j)\log\left[\frac{1 - PRj(s)}{1 - PRj(0)}\right]\right\}\end{aligned} \tag{10}$$

where $PRj(s)$ corresponds to the probability of observing X_j with $X_1 = s$.

In the special case where $PRj(X_1)$ corresponds to a logistic regression with main effects such that

$$\text{logit}(P(R_j = 1|X_1 = s, X_{(-1)})) = \phi_{0j}^{s} + \sum_{k=2, k\neq j}^{p} \phi_{kj}^{s} X_k$$

we have the following for $s > 0$:

$$\log\left[\frac{P(X_1 = s|X_{(-1)}, R)}{P(X_1 = 0|X_{(-1)}, R)}\right] = \theta_{0s} + \sum_{j=2}^{p} \theta_{js} X_j$$
$$+ \sum_{j=2}^{p}\left\{R_j[\phi^s_{0j} + \sum_{k=2,k\neq j}^{p} \phi^s_{kj} X_k] + \log[1 + \exp(\phi^s_{j0} + \sum_{k=2,k\neq j}^{p} \phi^s_{jk} X_k)]\right\} \tag{11}$$
$$- \sum_{j=2}^{p}\left\{R_j[\phi^0_{0j} + \sum_{k=2,k\neq j}^{p} \phi^0_{kj} X_k] + \log[1 + \exp(\phi^0_{j0} + \sum_{k=2,k\neq j}^{p} \phi^0_{jk} X_k)]\right\}$$

A first-order Taylor series approximation of equation (11) suggests a regression of the form:

$$\log\left[\frac{P(X_1 = s|X_{(-1)}, R)}{P(X_1 = 0|X_{(-1)}, R)}\right] \approx \omega_{0s} + \sum_{j=2}^{p} \omega_{js} X_j + \sum_{j=2}^{p} \omega^s_{Rj} R_j + \sum_{j=2}^{p} \sum_{k=2,k\neq j}^{p} \omega^s_{RjXk} R_j X_k \tag{12}$$

In other words, we can include the missingness indicators and their interactions with X as additional predictors. If we can further assume no interaction between X_1 and the other X's in the model for the missingness of X_j, then ϕ^s_{kj} takes a single value across $s = 1, \ldots, S$ for $k = 2, \ldots, p$, $k \neq j$. In this case, we have

$$\log\left[\frac{P(X_1 = s|X_{(-1)}, R)}{P(X_1 = 0|X_{(-1)}, R)}\right] \approx \alpha_{0s} + \sum_{j=2}^{p} \alpha_{js} X_j + \sum_{j=2}^{p} \alpha^s_{Rj} R_j \tag{13}$$

indicating that we should just include the missingness indicators in the imputation model.

For more general missingness mechanisms, we can apply a generalization of the offset strategy of equation (9) where we define offsets:

$$Z_{js} = R_j \log\left[\frac{PRj(s)}{PRj(0)}\right] + (1 - R_j)\log\left[\frac{1 - PRj(s)}{1 - PRj(0)}\right] \tag{14}$$

and impute from a regression model as follows:

$$\log\left[\frac{P(X_1 = s|X_{(-1)}, R)}{P(X_1 = 0|X_{(-1)}, R)}\right] = \omega_{0s} + \sum_{k=2}^{p} \omega_{ks} X_k + \sum_{k=2}^{p} Z_{ks} \tag{15}$$

where $\sum_{k=2}^{p} Z_{ks}$ is a fixed offset.

Imputing continuous variables

We now suppose that X_1 follows some continuous distribution defined on the real line. First, we will consider the special case where X_1 is normally distributed given $X_{(-1)}$. Then, we will propose a strategy for more general X_1.

Continuous Special Case 1: imputing normally distributed variable.

Suppose first that X_1 is normally distributed such that $X_1|X_{(-1)} \sim N(\theta_0 + \sum_{k=2}^{p} \theta_k X_k, \sigma^2)$. Suppose further that the probability of observing X_j is given by

$$\text{logit}\big[P(R_j = 1|X)\big] = \phi_{j0} + \sum_{k\neq j} \phi_{jk} X_k$$

Following equation (2), we can express the imputation model for X_1 as

$$f(X_1|X_{(-1)}, R_{(-1)}) \propto f(X_1|X_2, , X_p) \prod_{k=2}^{p} f(R_k|X_1, , X_p)$$
$$\propto \exp\left(-\frac{[X_1 - (\theta_0 + \sum_{k=2}^{p} \theta_k X_k)]^2}{2\sigma^2}\right) \times \prod_{k=2}^{p} \frac{\exp(R_k[\phi_{k0} + \sum_{s\neq k} \phi_{ks} X_s])}{1 + \exp(\phi_{k0} + \sum_{s\neq k} \phi_{ks} X_s)} \tag{16}$$

Consider the special case where $p = 3$ (so $X = (X_1, X_2, X_3)$). The two terms in equation (16) are, respectively, a bell-

shaped curve and the product of two separate bounded sigmoid functions as a function of X_1. The sigmoid curve for $f(R_2|X)$ will be increasing in X_1 for one value of R_2 and decreasing for the other value, and likewise $f(R_3|X)$ will be increasing in X_1 for one value of R_3 and decreasing for the other value. To represent a valid distribution, the product in equation (16) has to be normalized to integrate to 1. More generally, it is clear that the full conditional distribution of X_1 will depend on R_k, assuming $\phi_{k1} \neq 0$. Additionally, the conditional distribution of X_1 is *not* symmetric and its mean is no longer given by $\theta_0 + \sum_{k=2}^{p} \theta_k X_k$. While it is feasible to draw from the distribution proportional to equation (16) exactly, we will explore approximations that may be easier to draw from in practice.

Approximation Strategy 1: An intuitive approximation of equation (16) would be to draw X_1 from the following normal distribution:

$$N(\omega_0 + \sum_{k=2}^{p} \omega_k X_k + \sum_{k=2}^{p} \omega_{Rk} R_k, \tau^2) \tag{17}$$

This strategy can be justified as a second-order Taylor series approximation of equation (16) as follows. Assuming ϕ_{jk} is small for all k,

$$\log\left[f(R_j|X)\right] \approx R_j[\phi_{j0} + \sum_{k \neq j} \phi_{jk} X_k] + \log(1 + \exp(\phi_{j0}))$$
$$+ \frac{\exp(\phi_{j0})}{1 + \exp(\phi_{j0})}[\phi_{j1}X_1, , \phi_{jp}X_p]^T + \frac{\exp(\phi_{j0})}{\left[1 + \exp(\phi_{j0})\right]^2}[\phi_{j1}X_1, , \phi_{jp}X_p]^{\otimes 2}$$

where $\phi_{jj} = 0$. Combining these expressions with the form for $\log(f(X_1|X_{(-1)}))$ and collecting terms multiplied by X_1 in equation (16), we obtain a linear regression in the form of equation (17).

If the association between the X's and the R's is stronger, then this Taylor series approximation may be less accurate, and a more involved approach to drawing values of missing X's is needed. For example, we notice from equation (16) that X_2 appears in $f(X_1|X_{(-1)})$ and may also be included in the various missingness models, suggesting something more general than a linear term in X_2 may be needed for imputing X_1. We propose including a spline function of X_2. Similar spline terms could also be included for other covariates in the imputation model. This results in the following approximate imputation distribution:

$$N\left(s_2(X_2) + s_3(X_3) + \cdots + s_p(X_p) + \sum_{k=2}^{p} \omega_{Rk} R_k, \tau^2\right) \tag{18}$$

where $s_k(X_k)$ denotes a spline function of covariate X_k. The presence of the product of sigmoid curves in equation (16) modifies both the spread and skewness of the imputation distribution. We will ignore the skewness, but we could accommodate the spread by letting it depend on the values of R. Thus, another level of approximation would be drawing X_1 from a normal distribution

$$N\left(s_2(X_2) + s_3(X_3) + \cdots + s_p(X_p) + \sum_{k=2}^{p} \omega_{Rk} R_k, \tau^2_{(R2,R3,,Rp)}\right) \tag{19}$$

As an even more flexible approximation, we might allow the variance to depend on R and incorporate interactions between X and R in the mean structure of the imputation distribution. The approximations in equations (18) and (19) could be incorporated into a sequential regression multiple imputation procedure, provided the software being used had the ability to include splines instead of simple linear terms in the mean structure of the regression model. In practice, a large value of n may be required to actually fit the largest of the above models during the imputation procedure. To build in even more flexibility in the imputation model, we might take a generally more robust approach to multiple imputation, such as predictive mean matching,[15,16] or random forests[17] conditioning on $R_2, , R_p$ in addition to other variables when imputing X_1.

Approximation Strategy 2: Rather than approximating the mean structure of equation (16) using Taylor series approximations, we could instead consider the mode of the distribution in equation (16), which we call $mode(X_{(-1)}, R_{(-1)})$. Assuming the distribution in equation (16) is uni-modal, then we might impute missing X_1 from $N(mode(X_{(-1)}, R_{(-1)}), \tau^2)$. Taking the derivative with respect to X_1 of the log of equation (16) leads to the following expression:

$$-\frac{X_1 - (\theta_0 + \sum_{k=2}^{p} \theta_k X_k)}{\sigma^2} + \sum_{k=2}^{p} \phi_{k1}[R_k - PRk(X_1)] \tag{20}$$

where $PRk(X_1) = P(R_j = 1|X) = \text{expit}(\phi_{j0} + \sum_{k \neq j} \phi_{jk} X_k)$ is viewed as a function of X_1. Assuming a uni-modal distribution, we can obtain the $mode(X_{(-1)}, R_{(-1)})$ by setting equation (20) equal to 0 and solving for X_1. Finding this mode is numerically feasible, but the form of equation (20) suggests an alternative approach for imputing X_1 within the iterative imputation algorithm:

1. For each $j > 1$, fit a logistic regression model to the current imputed dataset of size n for the probability that X_j is observed.
2. Using the most recent imputed values for X_1 and the latest estimates of ϕ and $PRj(X_1)$ obtained in step 1, define new variables $Z_j = \phi_{j1}[R_j - PRj(X_1)]$ for each $j > 1$.
3. Impute X_1 using the following model:

$$N\left(\omega_0 + \sum_{k=2}^{p} \omega_k X_k + \sigma^2 \sum_{k=2}^{p} Z_k, \tau^2\right) \tag{21}$$

where the ω's are drawn from the approximation to their posterior distribution obtained by fitting equation (21) to the full imputed dataset and $\sigma^2 \sum_{k=2}^{p} Z_k$ is treated as an offset using the estimate of σ^2 obtained from fitting the model for X_1 given $X_2, , X_p$ to the complete data. Alternatively $\sum_{k=2}^{p} Z_k$ could be added as another predictor in the imputation model.

Continuous General Case: non-normal continuous variable.

Suppose X_1 takes a more general non-Gaussian continuous form and that $X_2, \ldots, X_p$ take unspecified forms. For this case we may first transform X_1 so that the conditional distribution of $X_1|X_2, \ldots, X_p$ is approximately Gaussian with constant variance σ^2. Using the intuition developed for normally distributed X_1 considered above, we propose the following three strategies for approximating the conditional imputation distribution for X_1 in equation (2) using one of the following three imputation distributions:

$$N\left(\omega_0 + \sum_{k=2}^{p} \omega_k X_k + \sum_{k=2}^{p} \omega_{Rk} R_k, \tau^2\right) \tag{22}$$

$$N\left(\sum_{k=2}^{p} s_k(X_k) + \sum_{k=2}^{p} \omega_{Rk} R_k, \tau^2\right) \tag{23}$$

where $s_k(X_k)$ is a spline function of X_k, and

$$N\left(\omega_0 + \sum_{k=2}^{p} \omega_k X_k + \sigma^2 \sum_{k=2}^{p} Z_k, \tau^2\right) \tag{24}$$

where $Z_k = \phi_{k1}[R_k - PRk(X_1)]$ is a constructed variable based on estimated probability of observing X_k, $PRk(X_1) = P(R_k = 1|X_{(-k)})$, obtained using the most recent imputed data.

Simulation studies

Simulation set-up

We performed numerical studies to investigate the performance of the proposed method under different missingness and X distribution settings. For each setting, we generate 500 simulated datasets with 2000 subjects each. In each simulated dataset, we generate 5 correlated variables under two different scenarios. In the first scenario, we simulated 5 multivariate normal variables $X_1, , X_5$ with mean 0, unit variances, and covariances $\Sigma_{jk} = cov(X_j, X_k)$ as follows: $\Sigma_{12} = 0.4$, $\Sigma_{14} = \Sigma_{35} = 0.3$, $\Sigma_{13} = \Sigma_{25} = \Sigma_{34} = 0.2$, and all remaining covariances equal to 0.1. In the second scenario, covariates X_1, X_2, and X_3 are dichotomized to take the value 1 if the drawn value is above zero. We then impose roughly 25-50% missingness in each of X_1, X_2, and X_3 under the following models:

$$\text{logit}(P(R_1 = 1|X_2, X_3, X_4, X_5) = \phi X_2 + \phi X_3 + \rho X_4 + \rho X_5$$
$$\text{logit}(P(R_2 = 1|X_1, X_3, X_4, X_5) = \phi X_1 + \phi X_3 + \rho X_4 + \rho X_5$$
$$\text{logit}(P(R_3 = 1|X_1, X_2, X_4, X_5) = \phi X_1 + \phi X_2 + \rho X_4 + \rho X_5$$

where $\phi = 0$, 0.25, 0.50, 0.75, 1, or 1.5 and ρ was either 0 or 1. Corresponding complete case probabilities ranged between 12% and 50%.

For each simulated dataset in each setting, we obtained 10 multiple imputations for missing values in X_1, X_2, and X_3 a subset of the following methods:

1. SRMI: usual chained equations assuming missing at random
2. SRMI-MI: method SRMI + adjusting for missingness indicators as in equations (5) and equation (17)
3. SRMI-Interactions R: method SRMI-MI + adjusting for missingness indicator-covariate interactions as in equations (7) and (19)
4. SRMI-Interactions X: method SRMI-MI + adjusting for covariate-covariate interactions as in equation (6)
5. SRMI-TriCube: adjusting for missingness indicators and cubic splines for other covariates as in equation (18)
6. SRMI-Offset(Normal): method SRMI + estimated offset as in equation (21)
7. SRMI-Offset(Binary): method SRMI + estimated offset as in equation (9)
8. SRMI-Exact: imputing from "exact" distribution proportional to equation (2), using drawn missingness model parameters

The SRMI-Exact method imputes missing values from the correct conditional distribution after estimating missingness model parameters in the observed data. This method serves as a benchmark for the various (more easily implemented) approximations considered.

For scenarios with normally distributed or binary X_1, X_2, and X_3, we performed imputation using a subset of the above methods relevant for the corresponding covariate distributions as motivated by our derivations above. For the SRMI-Offset(Normal), SRMI-Offset(Binary), and SRMI-Exact methods, we assumed a logistic regression model structure for missingness in each variable, and we estimate or draw corresponding missingness model parameters using the most recently imputed data. Parameters in the missingness model can be estimated well, as demonstrated by simulation Supplemental Figure A.1. For each simulation setting and imputation strategy combination, we obtained point estimates for (1) the mean of X_1 and for (2) regression coefficients from a model for $X_1|X_2, X_3, X_4, X_5$ using the multiply imputed data and Rubin's combining rules. We then calculated the average bias, empirical variance of the point estimates, and the coverage rate of 95% confidence intervals across the 500 simulated datasets. Results using a larger number of multiple imputations were similar.

Simulation results

Figure 1 shows the bias in estimating the mean of X_1 for different imputation methods. Shaded regions provide a visualization of the Monte Carlo standard error as discussed in Morris et al.[18] Under MAR ($\phi = 0$), none of the methods gave substantial bias. For both normally distributed and binary variables, SRMI produced substantial bias (e.g. absolute bias of 0.10 for normal X_1) under MNAR ($\phi \neq 0$). In both normal and binary settings, all MNAR adjustment methods considered resulted in similar or reduced bias relative to SRMI (e.g. SRMI-MI resulted in up to 80% reduction in bias relative to SRMI for normal X_1). The SRMI-MI method worked well to reduce bias from MNAR missingness when (1) MNAR missingness was weak or (2) missingness did not depend on the continuous variables ($\rho = 0$).

In the setting with very strong MNAR missingness or missingness dependent on continuous covariates, the SRMI-MI approximation resulted in large residual bias (e.g. absolute bias of -0.07). For imputation of normally distributed covariates, the SRMI-Exact method was the only approach that consistently produced good properties in terms of bias. Imputation models using more complicated functions of predictors (e.g. interactions, splines) often provided smaller bias relative to SRMI-MI but did not perform as well as imputation using the "exact" conditional distribution, particularly when $\rho \neq 0$. For imputation of binary covariates, the offset approach generally performed well in terms of bias reduction, particularly when missingness model parameters were fixed to the simulation truth (not shown). Some small residual bias was seen for the offset method when missingness model parameters were estimated. Although not shown, complete case analysis resulted in very large bias in all simulation settings considered. Biases for regression model coefficients are presented in Supplemental Figure A.3. Results are similar.

Figure 2 shows the empirical variance of point estimates for the mean of X_1, relative to analysis of the full data with no missingness. Under true MAR missingness, there is at most a small increase in the variability due to the extensions of the SRMI method relative to standard SRMI. Inclusion of additional interaction terms (between missingness indicators and covariates or between covariates themselves) in the imputation models resulted in larger empirical variances. In the setting with normally distributed covariates, SRMI-Exact imputation resulted in larger empirical variance when the MNAR missingness was very strong. However, coverage rates (Supplemental Figure A.2) were similar to other

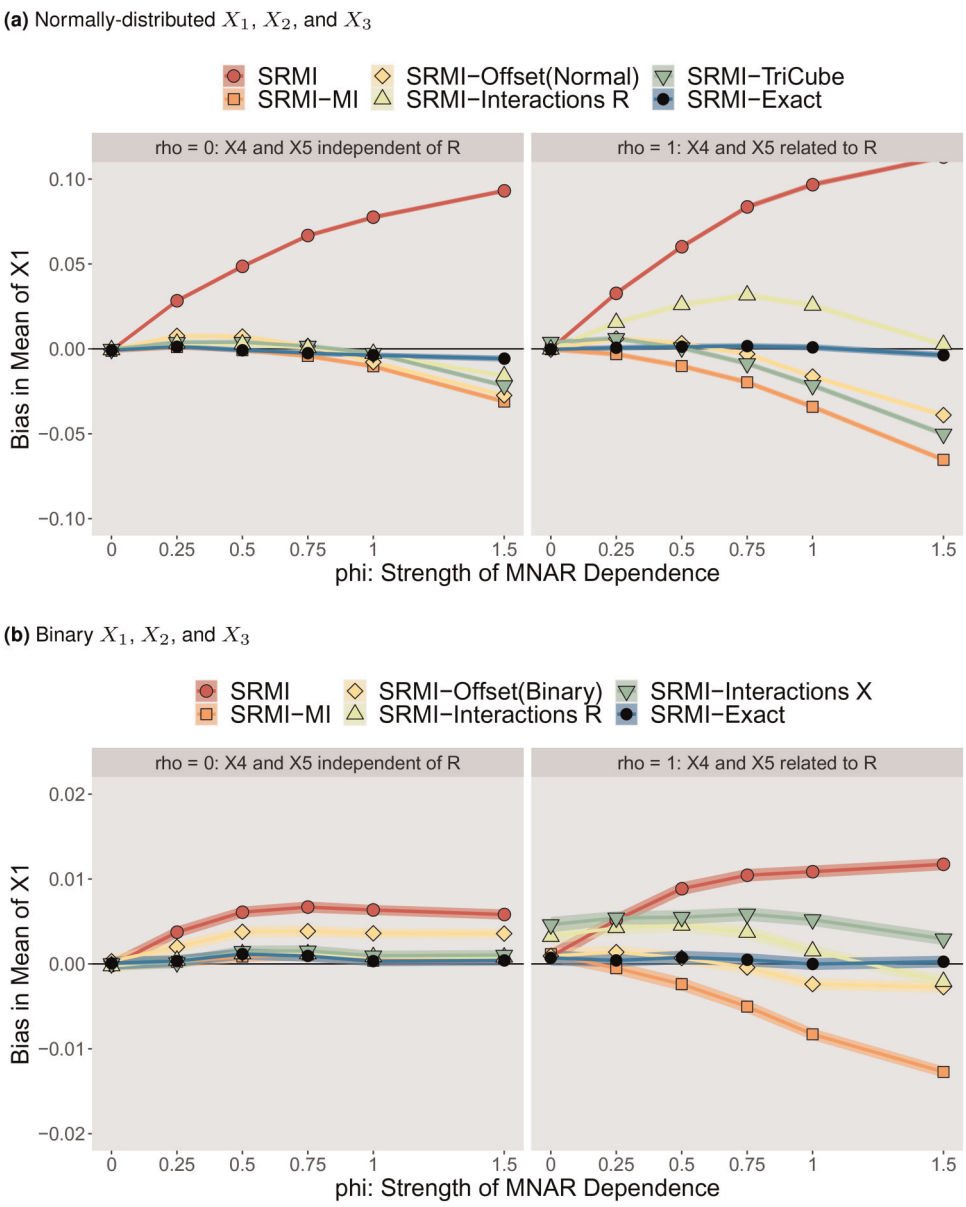

Figure 1. Bias for mean of X_1 across 500 simulated datasets after applying various imputation strategies. Results shown for M = 10 imputed datasets. Shaded regions correspond to Monte Carlo standard errors.

methods, indicating that there may be a trade-off between wider confidence intervals and lower bias when applying these methods to account for MNAR missingness.

Prevalence of genetic pathogenic variants in breast cancer patients

The methodological development in this paper was motivated by missing data challenges for the ICanCare study. This study consists of women aged 20–79 years who were newly diagnosed with breast cancer between July 2013 and August 2015 and are part of the Surveillance, Epidemiology, and End Results (SEER) registries in Georgia and Los Angeles. SEER is a population-based registry that collects basic data on variables such as age, race, stage of disease, common breast cancer biomarkers, and treatments. A subset of these women enrolled in the ICanCare study,[19] in which they were surveyed about the care they received and many other factors. The ICanCare study broadly focused on treatment communication and decision making in patients with favorable breast cancer. Women were also asked about whether they had a family history of breast cancer, and they provided other information related to their risk of

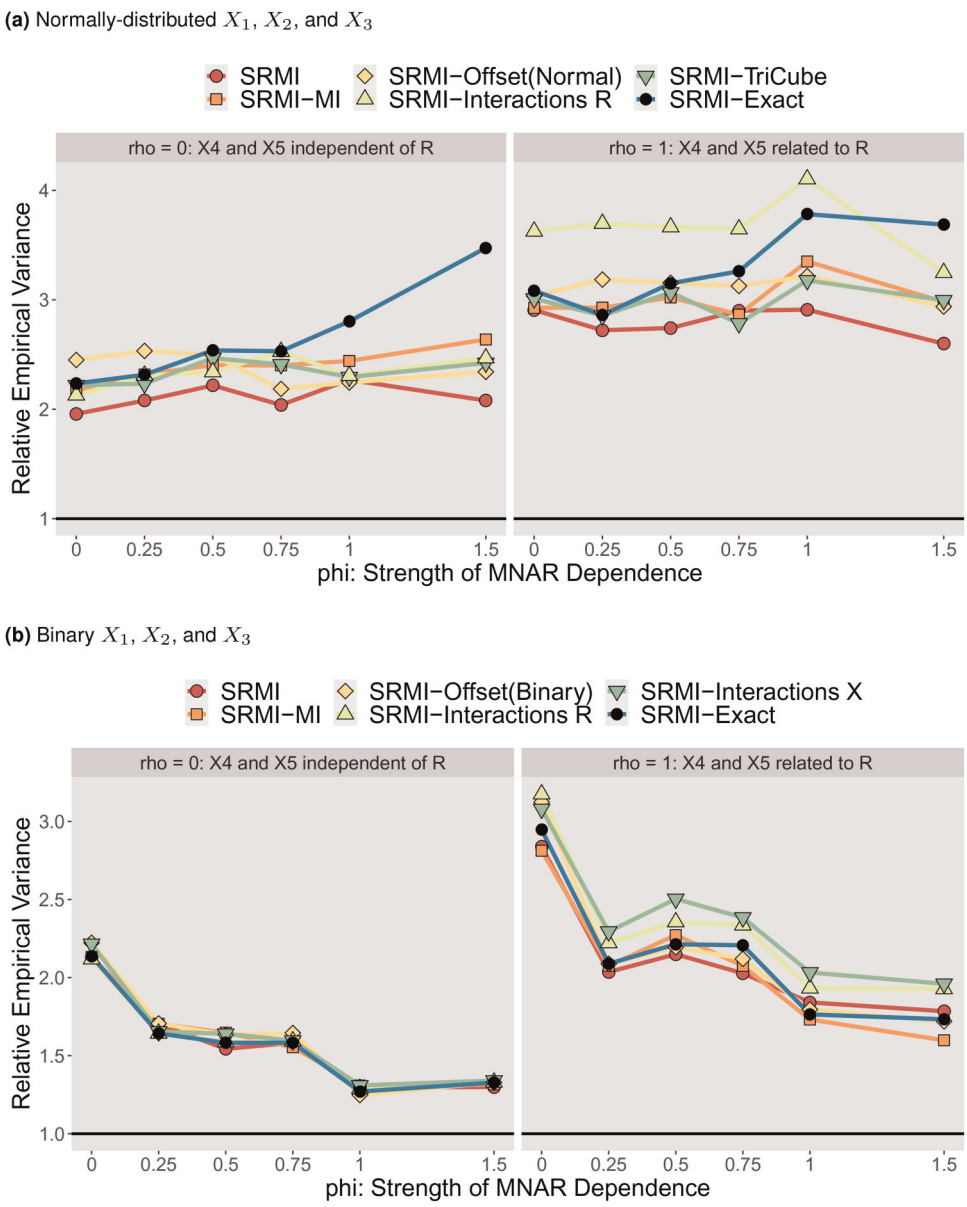

Figure 2. Relative variance for estimated mean of X_1 across 500 simulated datasets after applying various imputation strategies, relative to analysis of the full data with no missingness.[1] (a) Normally distributed X_1, X_2, and X_3 and (b) binary X_1, X_2, and X_3.[2]

being a carrier of genetic variants associated with breast cancer. The survey was completed by 5080 patients and linked to SEER data. In addition, genetic test results corresponding to pathogenic variants were available for some patients. An external company merged the survey responses and SEER clinical data with genetic testing information obtained from four laboratories that tested patients in the study regions and provided a de-identified dataset. More details regarding the combined datasets are provided elsewhere.[20]

In this paper, we are interested in using data from the ICanCare study to better understand the prevalence of the pathogenic genetic variants in BRCA1 or BRCA2 among women diagnosed with breast cancer in the USA. Women with breast cancer are increasingly taking genetic tests to find out if they have pathogenic variants in important genes. This information can impact the treatments they receive and is relevant for the care of close relatives. The most well-known breast cancer genes are BRCA1 and BRCA2. The prevalence of pathogenic variants in BRCA1 and BRCA2 in the general population is quite low, estimated to be roughly in the 0.2–0.3% range.[21] Estimates of prevalence of BRCA1/2 pathogenic variants among breast cancer patients vary from country to country (typically around 2–4%), but can exceed 20% among breast cancer patients with a positive familial history of breast cancer.[22–24] Given the practical importance of these genetic

variants to patient prognosis and treatment decision-making, there is a great need to better characterize the prevalence of these pathogenic variants in the population of women newly diagnosed with breast cancer in the USA. Missing data, however, presents a challenge.

Genetic test results (including presence/absence of BRCA1/2 mutation) are not available for some patients in the ICanCare study. Amongst the 5080 women 27.5% had genetic test results, and amongst those with genetic tests 4.66% had a pathogenic variant in either BRCA1 or BRCA2. The current recommendation for genetic testing is based on patient age, personal or family history of cancer, known genetic mutation in the family, and tumor characteristics, although there is substantial variability in how much these recommendations are being followed.[25] Even if genetic testing is offered, patient interest in undergoing genetic testing is influenced by factors such as age, race, education, and stage of disease.[26] The sample of women who do have genetic test data results within the ICanCare study are very unlikely to be representative of all the women in the ICanCare study or of the population of all women in these two SEER registries, and we expect the estimated prevalence of mutation among women with observed genetic test results to be an over-estimate. More sophisticated strategies are, therefore, needed to address the missing data.

Our strategy for handling missingness in BRCA mutation status is to use other available data to multiply impute BRCA status for women with missing values. Then, we can estimate the prevalence of BRCA mutation in the ICanCare study using the multiply imputed data. For the purposes of this paper we will consider a single variable of whether either BRCA1 or BRCA2 has a pathogenic variant. Some key variables that will help inform our imputation of BRCA mutation status are presence of familial history of BRCA mutation, Jewish ancestry, and familial history of breast cancer. Age, race, presence of ER/PR/HER2 mutations, tumor grade, clinical T-stage, presence of lymph node invasion, and presence of bilateral disease may also be informative. Most of these variables had low missingness rates (0–5%), but HER2 status, family history of cancer and known familial BRCA mutation had higher missingness rates (10–21%). Summary statistics for these variables along with their missingness rates are given in Supplemental Table C.1.

Standard multiple imputation methods require us to assume that missingness in BRCA mutation status is independent of unobserved information given the observed data. However, this may not be the case. In particular, we may believe that presence of familial history of mutation, familial history of breast cancer, and other variables may strongly impact choices for whether or not a woman undergoes genetic testing. Since these variables also are observed with missingness, the MAR assumption may likely be violated. We see evidence of this dependence in the data. Logistic regression modeling of whether a woman had an available BRCA1/2 test result using data for the 2863 patients with complete covariate information showed an association between missingness and age, race, familial history of either breast, ovarian cancer or sarcoma, familial history of BRCA1/2 mutations, Jewish ancestry, HER2 status, and geographic location. The odds ratios for this logistic regression model are presented in Supplemental Table C.2. Since these variables are related to missingness in BRCA mutation status and are also occasionally or even often missing themselves, missingness in BRCA status may likely be MNAR.

This MNAR mechanism has a potential to induce bias in resulting estimates of BRCA mutation rates, since these variables are also related to whether or not the BRCA mutation was present. In particular, we ran a logistic regression model on the 874 patients that received a BRCA1/2 test and had complete information for the clinical and demographic factors listed above. In this logistic regression we used the Firth correction to avoid quasi-separation due to the rare outcome. The following variables were clearly associated with having a BRCA1/2 pathogenic variant: age, relatives with history of either breast, ovarian cancer or sarcoma, relatives with known BRCA1/2 mutations and Jewish ancestry(borderline). The odds ratios for this logistic regression model are presented in the first column of Supplemental Table C.3. Many of the same variables that are associated with receiving a test are also associated with the positivity rate of the test. Since these variables also have missingness themselves, there is a need to carefully guard against bias due to the MNAR missingness in the multiple imputation process.

We performed sequential regression multiple imputation of the missing data using the *mice* program in R using several of the methods explored in this paper. There were four variables with missingness exceeding 10%: BRCA1/2 test results, family history, known pathogenic variant and HER2. For these four variables we created response indicators R_j and offset variables Z_j from equation (9), $j = 1, ..4$. Multiple imputations were generated using the following three methods: (1) standard SRMI, (2) SRMI-MI, and (3) SRMI-Offset. When using the SRMI-MI method we imputed each variable j in the dataset conditional on all other variables, and all of the above $R_{(-j)}$. When using the SRMI-Offset method we imputed BRCA1/2 test, family history, known pathogenic variant, and HER2 status conditional on all other variables and all of the above $Z_{(-j)}$. The rest of the variables were imputed conditional on all other variables and R_j.

Logistic regression models were used for imputing binary variables, and multinomial logistic regression was used for imputing variables with more than two categories. We treated clinical stage and tumor grade as categorical. For binary variables with low prevalence, imputation using the "logreg" option in *mice* is unstable and can produce bias in downstream prevalence estimates as shown by simulation in Supplemental Materials. Although the main goal of this analysis

is to address potential MNAR missingness, this secondary problem posed a challenge for implementation of the methods proposed in this paper. In Supplemental Section B, we describe a modified strategy for drawing parameters for any imputation model structure that has better performance in the setting of rare binary outcomes. Imputation then proceeds using the proposed methods as described previously. We applied this method to impute BRCA1/2 status. For each imputation method, we obtain 10 multiple imputations based on sequential regression algorithms that were run for 50 iterations. The marginal prevalence of BRCA1/2 mutation was then estimated, along with corresponding standard errors.

Table 1 shows the estimated prevalence of BRCA1/2 pathogenic variants from the complete cases and from the different multiple imputation methods. As expected, the multiple imputation methods give lower estimates than the complete case analysis. The extensions of the SRMI that make use of the missing data indicators give slightly lower estimates than obtained from SRMI. Since we believe missingness is MNAR, we would trust results from the SRMI-MI and SRMI-Offset methods over the estimates from SRMI. In Supplementai Table C.3, we also present the estimated associations between having a BRCA1/2 pathogenic variant and the various risk factors for each of the imputation strategies. The results compared to the complete case analysis are broadly similar, but there are some differences. Notably, the associations for tumor grade and clinical T-stage are larger than in complete case analysis, and the associations for Jewish ancestry are smaller. As expected, the width of the 95% confidence intervals for the odds ratios from the multiply imputed datasets tend to be smaller than seen in complete case analysis.

Discussion

Standard software for implementing sequential multiple regression imputation (SRMI) assumes that missingness does not depend on unobserved information, called missing at random (MAR). Several researchers have proposed adaptations of existing sequential multiple imputation procedures in settings where missingness is not at random (MNAR).[9,27] For example, Tompsett and White proposes handling MNAR missingness by including missing data indicators as predictors in the sequential imputation models. In terms of rigorous statistical justification, however, little work has been done to provide guidance for handling of MNAR missingness within chained equations imputation algorithms in general.

In this paper, we provide statistical justification for the missing data indicators method of Tompsett and White and propose several extensions that can result in improved performance in terms of bias in the final data analysis. We approach this problem by first deriving the ideal imputation distribution as a function of observed data and assumed models for data missingness, viewing SRMI as an approximation to Bayesian MCMC estimation. Using Taylor series approximations and other methods, we obtain regression model approximations to the ideal imputation distribution to use in practice. We focus our attention on a particular MNAR setting, where missingness for a given variable may depend on *other* variables with missingness. The methods in this paper are not intended to apply to MNAR situations where there is good reason to believe the probability of missingness for a variable depends on the missing value of that variable. SRMI-MI is likely to give biased estimates in that setting, and the magnitude of the bias will depend on the strength of the associations in the missingness model and the strength of the associations between all the variables. It is plausible that SRMI-MI would work better than SRMI in many but not all such situations, and evaluation of the relative performance in the setting where missingness depends on the missing variable itself is beyond the scope of the current paper. We refer the reader to Beesley and Taylor (2021) for recent work addressing missingness in a given variable based on its *own* missing values.[14]

Through simulation, we found that inclusion of missingness indicators within sequential imputation algorithms (here, called SRMI-MI) can result in reduced bias in estimating outcome models parameters when missingness is MNAR following Assumptions 1 and 2. The degree of bias reduction will likely depend on the strength of the MNAR missingness and the structure of the missingness model. Although not explored here, inclusion of extra parameters in the imputation models could increase the risk of overfitting and may require larger datasets in order to see good bias reduction properties. In our simulations (datasets of size $n = 2000$), we did not see increase in bias or substantial increases in variance when SRMI-MI was applied instead of SRMI when missingness was truly MAR.

Table 1. Estimated prevalence of BRCA1/2 pathogenic variants.

	Estimate (100×)	Standard error (100×)
Complete cases	4.66	0.56
SRMI	2.82	0.47
SRMI-MI	2.77	0.39
SRMI-Offset	2.65	0.34

In some settings, SRMI-MI produced substantial residual bias. We proposed a variety of extensions to the SRMI-MI approach, including use of spline functions of model predictors, inclusion of interactions, and use of fixed offsets calculated as a function of estimated missingness model parameters. In general, approaches including additional interaction terms tended to result in increased standard errors with some benefit in terms of bias reduction. Of all the regression model approximations, the approach using missingness model-based offsets had the best properties on average across the many simulation settings considered. This may be because this approach is making use of more information from the data, since it involves assuming (and fitting) a model for the probability of missingness for each variable. Since we assume missingness in a given variable is independent of its own missing values, parameters in this missingness model may be identified using the observed data. However, this approach may be more sensitive to misspecification of the missingness model.

For comparison, we evaluate the performance of the various SRMI adaptations to imputation using the "exact" imputation model in equation (2). This distribution may only be known up to proportionality, and imputation using this distribution may be complicated in general. In our simulations, this approach (SRMI-Exact) resulted in little or no bias in estimating outcome model parameters.

With the exception of the SRMI-Exact method, we tried to restrict our focus to methods that are easily implemented within established sequential imputation software. The methods using the offset do require some non-trivial adaptations of the standard SRMI routine (including fitting of models for covariate missingness within the iterative imputation algorithm), and we provide example code guiding implementation with package *mice* in R.

The methods were applied to address potential MNAR missingness in data from the ICanCare study, which consists of a probability-sampled cohort of breast cancer patients identified from two SEER registries.[19] Sampling weights and sampling design information for the ICanCare study could be incorporated into imputation and analysis in order to generalize results to the entire SEER registry.e.g., [28–30] In a naive exploration into selection bias adjustment, we performed analysis weighted by the provided sampling weights but *ignoring* these weights during imputation. The estimated weighted prevalence of a pathogenic variant of BRCA1/2 obtained after SRMI-MI imputation was 2.55%. The corresponding unweighted estimate was 2.65%. Future efforts will implement more sophisticated strategies in the survey literature to simultaneously account for selection bias and MNAR covariate missingness when obtaining multiple imputations for these data.

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Supplemental Material

Supplemental material for this article is available online.

Notes

1. Results shown for $M = 10$ imputed datasets. Shaded regions correspond to Monte Carlo standard errors.
2. Results shown for $M = 10$ imputed datasets.

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